

Graph Class - 4

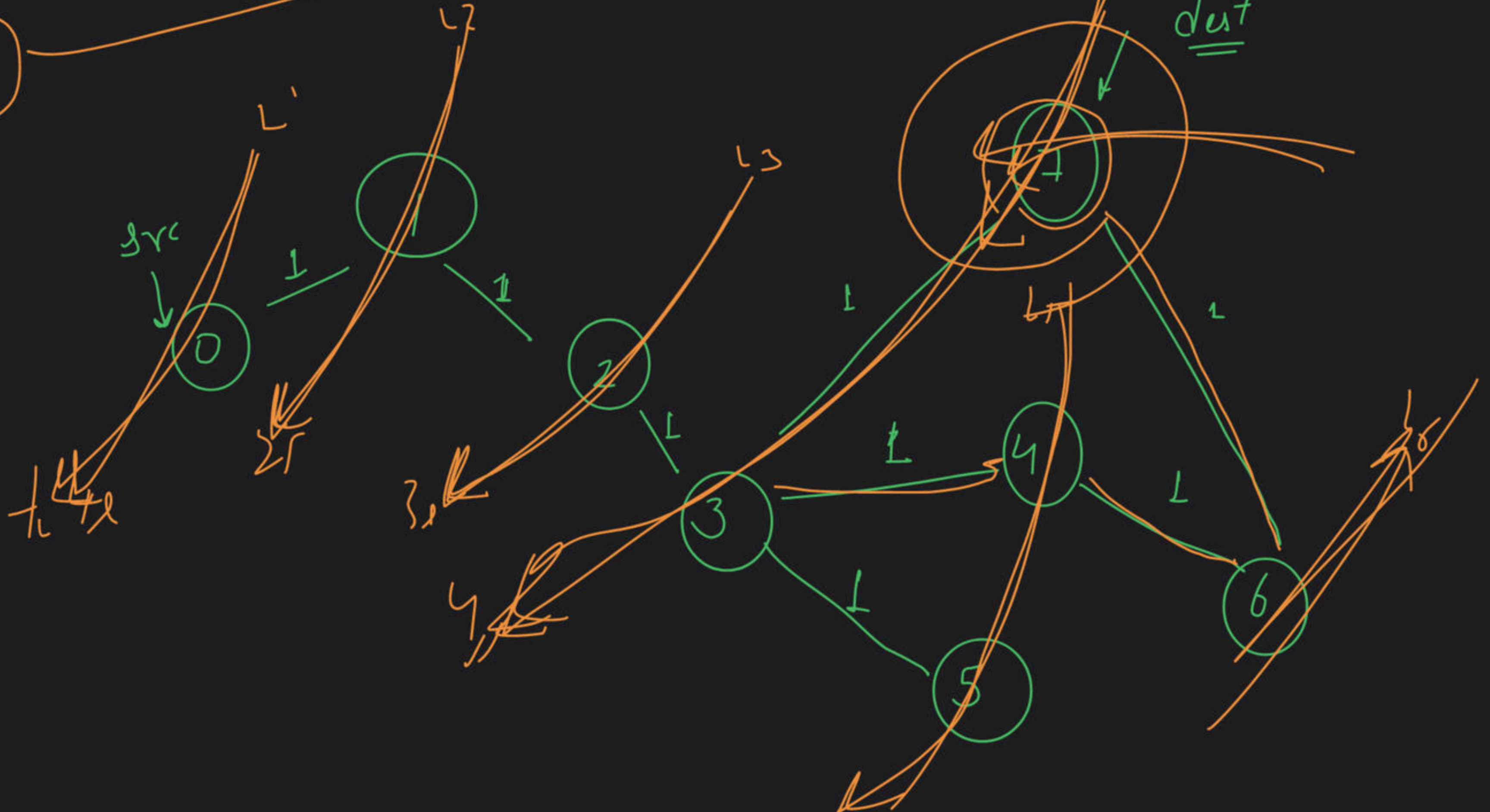
Special class

→ Shortest path

Observation - first time travel/unit → shortest path

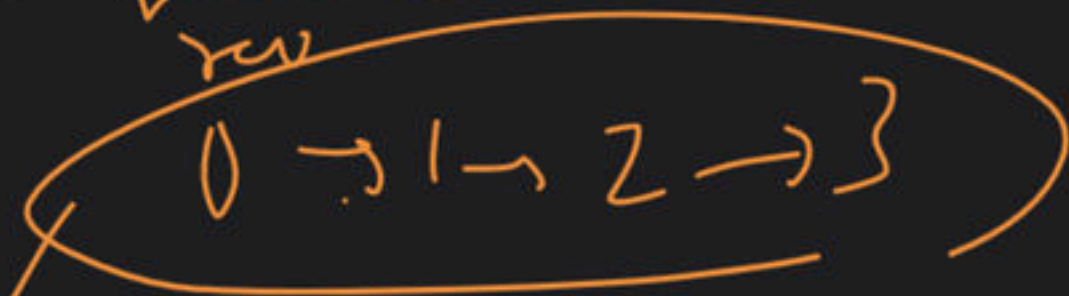
① BFS

SSSP

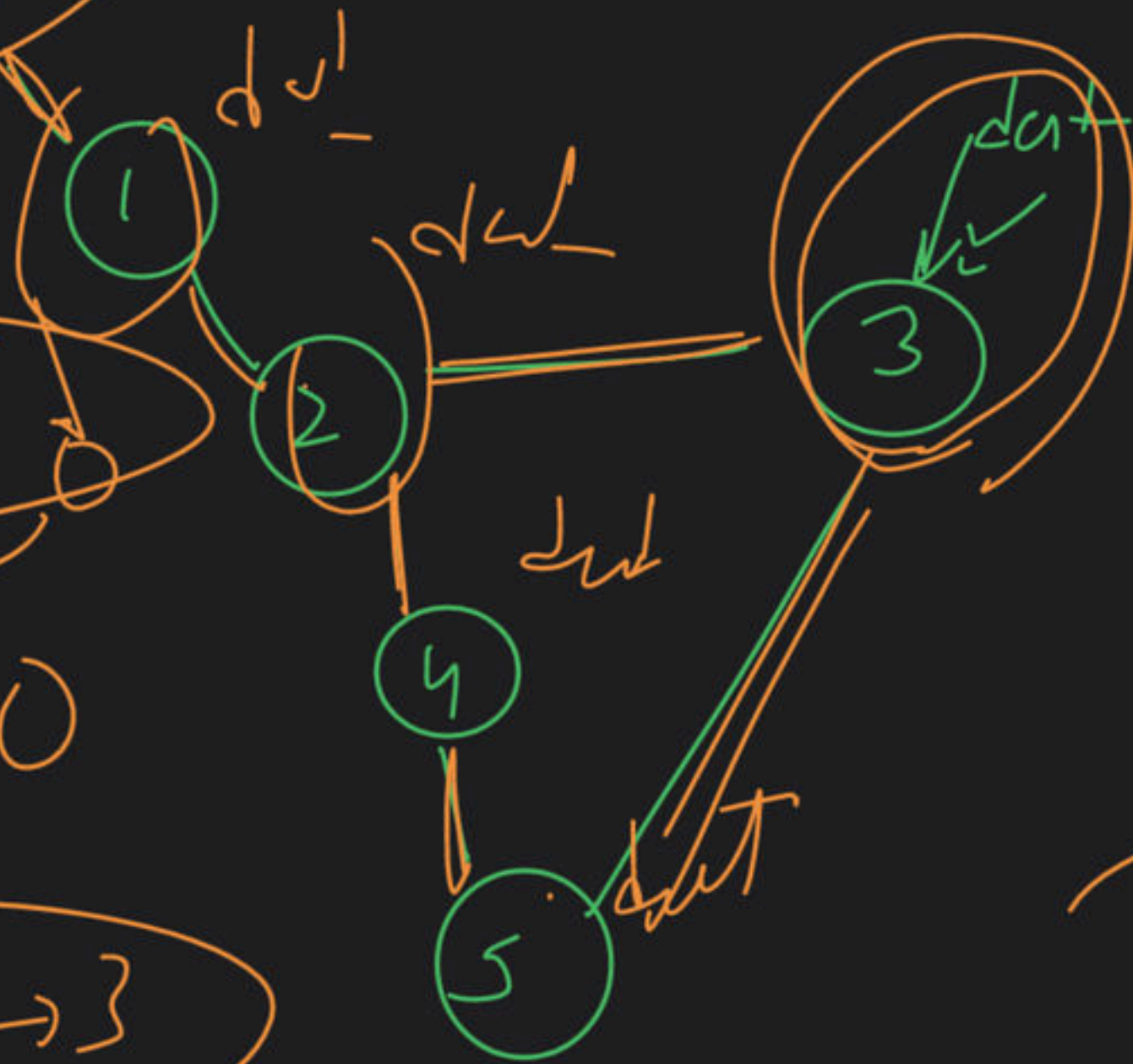




3 → 2 → 1 → 0



queue



Parent

0	→	-1
1	→	0
2	→	1
3	→	2
4	→	2
5	→	3

visited

0	→	T
1	→	T
2	→	I
3	→	T
4	→	T
5	→	I

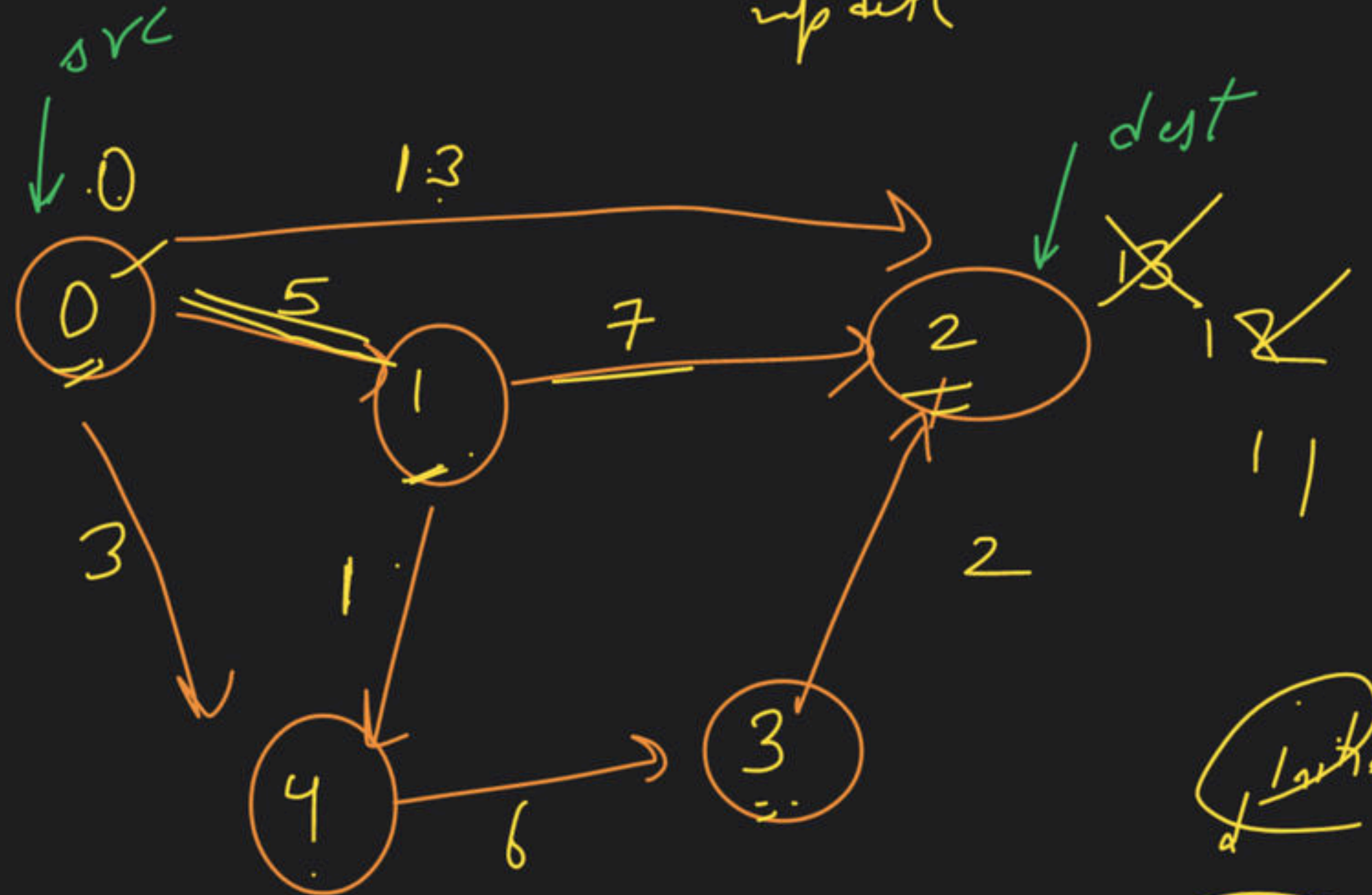
→ Directed graph → SSSP
 ↳ DFS

[min distance]
 ↳ best shortest distance
 update

5 examples

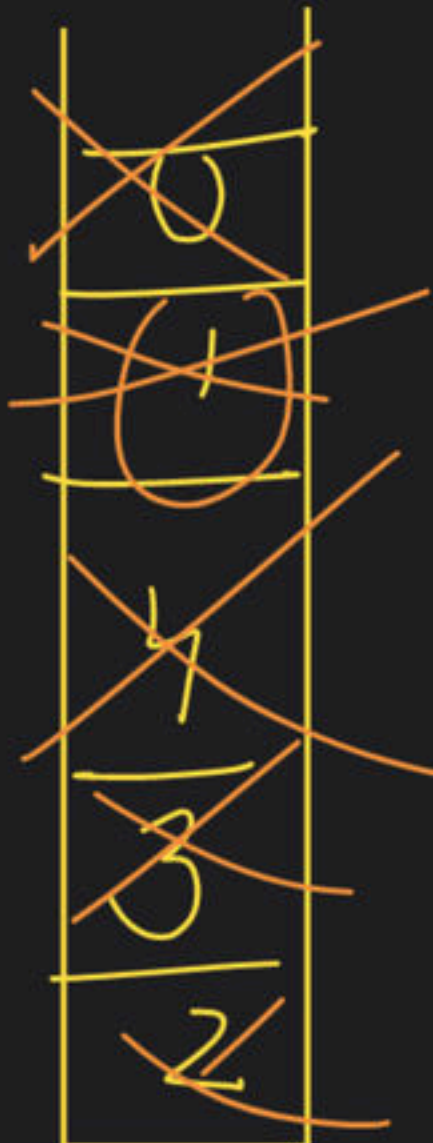


2 → ∞



Initial
 $dist(src) = 0$

stack



	0	1	2	3	4
dist	0	5	11 ∞	9	3

Kal



8:30 → 9:00

















































